

Raw SERS data



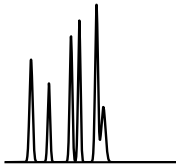
SERS map

one spectrum
per pixel



NNLS gate

background
dropped

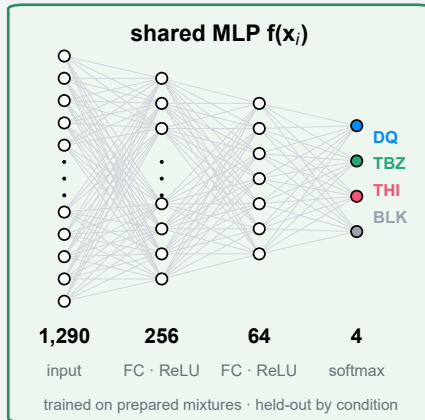


pixel spectra x_i

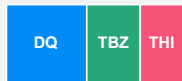
1,290 pts · 500–2500 cm^{-1}

Compositional unmixing

output



map composition



gated pixels · composition

$\bar{p}_i = \text{pixel mean} \cdot \text{BLK removed}$

absent component → reported 0

(absence gate, $P < 0.2$)

Concentration

output

reported concentration

1. correction net · 3–24 μM

2. pixel-library k-NN

surface μM ; dispensed in-window

outside library → no answer

declared total (optional)

$$\hat{C}_i = \bar{p}_i \cdot C_{total}$$

overlay when formulation known

guards: library distance > 3 → refuse · above window → ceiling only